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Desalination with Solid-Waste Valorization: A Sustainable Approach to Produced Formation Water Management

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Abstract

Produced formation water (PFW), with its high salinity and complex composition, poses a persistent challenge in petroleum operations, particularly in aging oilfields. Conventional disposal methods face increasing environmental and regulatory pressures. This paper presents an innovative approach adopted by Oil India Limited (OIL) that integrates reverse osmosis (RO)-based desalination with solid-waste valorization, transforming this waste stream into dual revenue streams - industrial reusable water and saleable solid by-products.

In OIL's oilfields in Dibrugarh and Tinsukia (Assam), a pilot project implemented a Zero Liquid Discharge (ZLD) RO-based desalination system. Effluent Treatment Plant (ETP)-treated produced water was processed through Low-Pressure RO (LPRO) followed by High-Pressure RO (HPRO). The reject from HPRO was concentrated via an electric waste heat evaporator and finally dried using an agitated thin film dryer. This process yielded reusable water, minimal vapor loss (0.2%), and solid waste. The system was tested at the Secondary Tank Farm (STF)-Madhuban in Duliajan, Assam, targeting sustainable inland disposal or reuse while exploring potential reuse of solid by-products.

The pilot demonstrated a water recovery efficiency of 99.975%, producing water that met all 21 discharge standards of the Environment (Protection) Act, 1986, enabling its reuse or safe inland disposal. Solid residues, primarily Halite and Natrite, posed an initial disposal challenge. However, OIL undertook R&D to explore value-added applications of these residues. Initial findings showed promising results to convert the solids into usable materials, such as fly-ash based construction materials. This initiative supports a circular economy by minimizing waste and creating marketable outputs from previously discarded materials. The integration of RO-based desalination and solid-waste valorization transforms produced water from a liability into a resource. Furthermore, the approach is scalable and adaptable for offshore platforms where the desalinated water could be reused onboard, and solids repurposed eliminating the need for reject discharge into marine ecosystems. This strategy addresses environmental concerns, enhances sustainability in hydrocarbon operations, and opens new revenue streams from waste valorization.

This paper introduces a pioneering dual-benefit strategy combining desalination with solid-waste valorization in produced water management a first of its kind in Indian upstream oil operations. It not only addresses environmental compliance and operational efficiency but also demonstrates techno-commercial

feasibility. By presenting an integrated, scalable, and replicable model for offshore and onshore use, the paper adds significantly to the state of knowledge in sustainable water reuse and waste-to-value practices in the energy sector.

keywords: Produced Formation Water, Desalination, Solid Waste Management, Water Reuse, Technology Development

Introduction

Crude oil reservoirs are naturally accompanied by associated formation water and gas. As reservoirs age and enter later stages of production, the ratio of water to oil, commonly referred to as the water cut, increases substantially (Lim et al., 2015). This rising volume of PFW represents a significant operational and environmental challenge for the oil and gas industry, not only due to its volume but also because of its high salinity and complex chemical composition.

In the mature oilfields of Assam operated by Oil India Limited (OIL), produced formation water (PFW) management has become increasingly challenging. With average PFW production at 5,000 kLpd, rising water cuts highlight the urgent need for advanced strategies. ETP treated PFW often retains high TDS and chloride levels, rendering it non-compliant for inland discharge or reuse. Regional factors such as Assam's high rainfall, make surface evaporation impractical, while the absence of marine access has led OIL to rely on reinjection into disposal wells or oil-bearing formations for EOR. Limited reuse in well-servicing operations exists but handles low volumes. These methods, though functional, face growing constraints due to technical, environmental, and economic challenges.

To address this, OIL piloted a RO-based desalination unit under a Zero Liquid Discharge (ZLD) model. It achieved high recovery of water meeting all 21 environmental parameters for discharge/reuse, but also generated solid waste, primarily Halite and Natrite. With no regional Treatment Storage and Disposal Facilities (TSDF), OIL initiated R&D to explore value-added, techno-commercial uses for these salts, showing promising conversion into usable materials.

This paper examines the broader implications of these innovations, advocating a sustainable approach to PFW management, transforming environmental liabilities into industrial assets. It outlines transitional, scalable pathways for water stewardship in the hydrocarbon sector. Specific proprietary details of the methodology are not disclosed.

Background

In hydrocarbon reservoirs, particularly oil reservoirs, the presence of PFW is a natural occurrence. Oil reservoirs typically contain a larger volume of water due to the lower compressibility and higher sorption capacity of liquids in comparison to gas reservoirs which have low porosity and are associated with low water capacity source rocks. (Guerra et al., 2011).

A simplified diagram of a typical oil reservoir (see Fig. 1) illustrates the stratified arrangement of gas, oil, and water layers structured by their respective densities, influenced by gravitational forces. The water, being the densest, settles at the bottom of the formation and is known as formation water. While underground, it is often termed as connate water, however, once brought to the surface during hydrocarbon production, it is commonly referred to as PFW (Igunnu and Chen, 2014). The generated volume of crude oil with respect to PFW depends on location and reservoir age (Abdulredha et al., 2020). Generally, it is accepted that for every unit of production of oil, three units of PFW is produced (Fakhru'l-Razi et al., 2009).

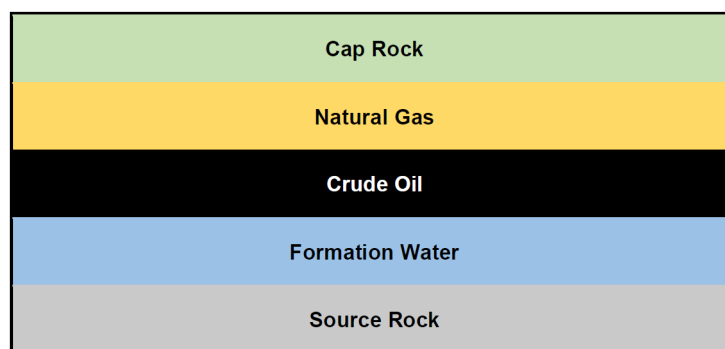


Figure 1—A schematic representation of a petroleum reservoir depicting the stratified layers.

Characterization of Produced Formation Water

Comprehensive characterization of PFW is critical for accurate identification and quantification of the present constituent composition, which is complex and highly variable due to location, reservoir formation and age, depth, production methods as well as the type of hydrocarbon being produced (Al-Ghouti et al., 2019). Proper characterization informs the selection and design of effective treatment technologies, helps evaluate treatment performance, and supports environmental and public health risk assessments while ensuring compliance with national regulatory standards.

Generally, PFW constituents can be classified into three categories as physiochemical, organic and inorganic, based on relevance to treatment behavior and operational impact. Physicochemical constituents include pH, turbidity, salinity, total dissolved solids (TDS), electrical conductivity, suspended solids, dissolved oxygen, and oil & grease, etc. These influence treatment selection, fouling and scaling tendencies, corrosion risk, and overall system performance. Organic constituents include benzene, toluene, ethylbenzene, and xylene (BTEX) compounds, polycyclic aromatic hydrocarbons (PAHs), phenols, carboxylic and fatty acids, surfactants, and light hydrocarbons. These compounds contribute to chemical oxygen demand (COD), emulsification, biological activity, membrane fouling, and environmental toxicity. Inorganic constituents comprise of major cations (Na^+ , Ca^{2+} , Mg^{2+} , K^+) and anions (Cl^- , SO_4^{2-} , HCO_3^-), as well as transition metals, heavy metals (e.g., Pb, Cr, As, Cd), alkaline earths (e.g., Ba, Sr), and other trace elements. These impact salinity, scaling potential, corrosion, and regulatory compliance.

Impact Analysis of Produced Formation Water

Environmental concerns from PFW disposal were reported since the inception of the oil and gas industry in the 19th century (Habashi, 2000). The impact of these pollutants is not uniform and depends heavily on both their concentration and the characteristics of the receiving environment. High-volume discharges in confined or low-flow water bodies tend to result in more acute ecological damage due to reduced dilution capacity. In elevated concentrations, these constituents raise risks on environment and public health. (refer Table 1).

Table 1—Major pollutants in PFW, their sources, and associated concerns.

Pollutant	Source	Concern	Reference
Cation Na ⁺	Rock-water interactions	Soil salinization, poor soil structure, impaired plant nutrient uptake, groundwater contamination	Davis et al., 2007
Hydrocarbons (HCs)	Dissolved aromatic and phenolic fractions	Toxic to aquatic life, short-term disruption	Frost et al., 1998
BTEX & PAHs	Residual petroleum	Impacts food chain via bioaccumulation	Neff, 2002; Ravndal, 2011
Divalent cations (Ca ²⁺ , Mg ²⁺ , Ba ²⁺ , Sr ²⁺)	Naturally occurring in reservoir rock	Scale formation in pipelines and equipment, clogging of reinjection wells	Chen et al., 2019
Anions (Cl ⁻ , HCO ₃ ⁻)	Chemical composition	Increase in conductivity, corrosive and insoluble scale formation	Glass and Buenfeld, 2000; t Li, 2013
Sulfates (SO ₄ ²⁻)	Present in formation brine	Promotes H ₂ S production via sulphate-reducing bacteria, causes corrosion	Rajbongshi and Gogoi, 2021

Regulatory Compliance

In OIL, the management of PFW is regulated as per standards defined in Schedule VI of the Environment (Protection) Act, 1986 and are set by the Central Pollution Control Board (CPCB) of India (refer [Table 2](#)). These rules mandate compliance with specific limits for 21 key parameters to mitigate environmental and public health risks (refer [Table 1](#)).

Pre-Treatment Analysis. The pre-treatment characterization of PFW involves evaluating the contaminant load to assess its treatability. As seen in [Table 2](#), there are high levels of oil and grease, TSS, TDS, chloride, and elevated BOD and COD, which are beyond the permissible limits defined by CPCB norms. These results show the complex and variable nature of untreated PFW, necessitating a robust and multi-stage treatment strategy.

Effluent Treatment Plant for PFW. Conforming to the analysis, the ETP design follows a systematic approach to pollutant removal, starting with oil-water separation and followed by chemical conditioning and solids filtration. The stages include the Corrugated Tilted Plate Interceptor (CTPI) for gravity-based separation, Dissolved Air Flotation (DAF) units for handling oils and suspended solids, and a combination of Multigrade Filter (MGF) and Oil Removal Filter (ORF) units for further polishing. Ultra-Filtration (UF) completes the process by removing residual colloidal and dissolved impurities.

Post-Treatment Analysis. As presented in [Table 2](#), there is a significant reduction in the levels of pollutants. TSS, BOD and COD, and oil & grease concentrations fall within acceptable limits. However, parameters such as TDS and chlorides remain above CPCB onshore discharge limits, highlighting the limitations of conventional treatment processes.

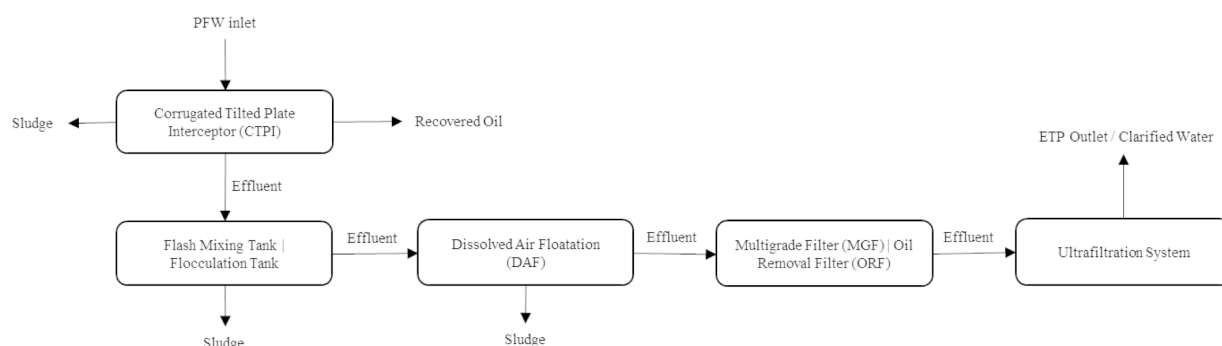


Figure 2—A schematic representation of ETP Process – Overview

Conventional Management of Produced Formation Water

Handling of PFW remains a critical operational challenge in the oil and gas industry. Due to the high salinity, complex composition and variable quality, appropriate handling and disposal methods must account for both environmental safeguards and operational feasibility. Below is a review of common existing methods for managing PFW, each influenced by regulatory requirements, water composition, volume handling needs, geographical constraints, and climate conditions.

Internal Reuse. While being location-independent and suitable across all climates, this method is requirement specific. It typically handles low to moderate volumes and may require adequate processing to meet operation and equipment compatibility. This method is currently operational within OIL for well-related services as a killing fluid.

Reinjection for EOR. This method helps maintain reservoir pressure and improves oil recovery by reinjecting treated PFW into hydrocarbon bearing formations. It is effective for managing moderate to high volumes and is climate independent. Feasibility depends on formation compatibility of the injected fluid, also it must meet specific standards to prevent scaling and formation damage. Regulatory compliance is the same as subsurface disposal which is satisfied for ETP treated PFW. Therefore, OIL actively uses this method in mature fields with existing reinjection infrastructure, making it a key part of its water management approach.

Subsurface Disposal. Subsurface disposal involves injecting treated PFW into deep underground formations via designated disposal wells. As it is independent of climatic and geographic constraints, this method is widely used in landlocked oilfields lacking surface or marine discharge options. Regulatory compliance is relatively straightforward, provided well integrity is ensured and injection occurs beyond the mandated depth of 1,000 meters (refer Table 2). OIL exceeds this requirement, injecting at depths of ~1,400 meters as an added safety measure.

While subsurface disposal accommodates high-salinity, high TDS water with less stringent pre-treatment, these characteristics also introduce challenges. Salts, divalent cations, and other scale-forming substances cause formation plugging and damage, requiring frequent well acidization. These interventions increase operational costs and downtime. Maintaining well integrity and monitoring groundwater aquifers are essential, as leaks or contaminant migration pose significant environmental and public health risks.

Despite its limitations, the upstream industry remains reliant on this method for its regulatory acceptability and ability to handle large volumes with minimal surface impact. However, its long-term sustainability is uncertain without stringent oversight, ongoing maintenance, and the development of alternative, more sustainable treatment solutions.

Surface Discharge. This disposal method is strictly regulated under the Environment (Protection) Act, 1986, enforced by the CPCB in India. It sets 21 key parameters to ensure safe discharge into inland surface waters. While feasible in areas with natural discharge points and capable of handling moderate volumes,

the treated effluent must consistently meet all prescribed limits. In OIL's case, although ETP processing reduces several pollutants, the water still exceeds permissible TDS and chloride levels (refer [Table 2](#)). As a result, this method is not currently used in OIL's water management strategy. While RO effectively reduces TDS and chlorides, it produces a concentrated reject stream of the same PFW.

Marine Disposal. Marine disposal of treated PFW is commonly employed in coastal oilfield operations due to its ability to handle large volumes. This method is resilient to climatic fluctuations and is favored where access to nearby marine bodies is available. However, it necessitates adequate treatment of the effluent to meet coastal discharge norms as per regulations.

Although treated water from ETP can meet all the required parameters for marine disposal (refer [Table 2](#)), this method remains inapplicable for OIL's operational areas, which are entirely landlocked in Assam with the nearest marine access being greater than 600 km. Consequently, marine disposal is not considered in OIL's produced water management strategy.

Evaporation ponds. These are increasingly discouraged by regulators due to their tendency to accumulate pollutants and are ineffective in high rainfall areas. These systems require vast open land and are not effective for large-scale PFW volumes. According to Assam State Disaster Management Authority (ASDMA), Assam has a high annual rainfall of 2900mm due to which, OIL does not employ this method.

Novel Methodology

Sustainable Management of Produced Formation Water

PFW management demands a transition toward sustainable practices, as conventional methods are increasingly constrained by environmental, regulatory, and operational limitations discussed earlier. Reinjection, which is highly adopted, also poses risks of formation damage, scaling, and high operational costs. Subsurface disposal of PFW represents the visible loss of a potentially recoverable resource for reuse. These shortcomings call for a sustainable approach that not only ensures regulatory compliance and long-term water stewardship but also facilitates value recovery from waste streams.

Desalination and Process Selection. PFW is not a universal commodity as its composition and properties vary across reservoirs. The results obtained with a desalination plant at one location may not be replicated at another location. With these concurrent limitations, OIL deployed a customized pilot trial of ZLD based desalination system (refer [Fig.3](#)) of capacity 2 m³/hr at Secondary Tank Farm (STF), Madhuban, Duliajan, Assam. The system primarily incorporated membrane-based separation and subsequent thermal concentration processes to maximize water recovery that meets the regulatory norms for inland discharge or internal re-use and isolate solid residues. The process was designed with a balance of proven efficacy, energy efficiency, operational feasibility and scalability.

PFW desalination typically results in a waste stream of either a brine concentrate or a solid extract, depending on the treatment architecture. The brine-based route merely shifts the disposal problem by concentrating pollutants, thereby posing the same challenges as conventional methods but amplified. In contrast, a ZLD approach yields only reusable water and a solid extract for further R&D, and thereby, was selected for its regulatory, operational advantages as well as to achieve the goal of sustainability ([Heins and Schooley, 2004](#)).



Figure 3—Full view of Pilot Desalination Plant.

ZLD Desalination Process Architecture

Feed PFW to Desalination System. The PFW generated at STF-Madhuban originates from the oil fields of Dibrugarh & Tinsukia districts of Assam. PFW is treated in ETP to meet the regulatory standards for sub-surface disposal as well as reinjection for EOR, or reuse as brine solution in well servicing. Primary treatment of PFW occurs at ETP through physical separation, chemical separation, and filtration as discussed earlier (refer Fig. 2). The PFW at ETP outlet (refer Table 2) was fed into the ZLD Desalination System.

Pre-treatment. Pre-treatment of the feed to the system involves coarse filtration units and fine filtration units. The coarse filtration unit contributed to removing larger particles from the feed water like larger suspended solids, particulate matter, etc. to protect the downstream system from clogging or damage. Likewise, the fine filtration unit contributed to removing very small particles in the range of 1 to 10 microns. The pre-treatment of feed PFW was carried out to protect RO-membrane from fouling and damage.

Reverse Osmosis. The PFW feed had TDS values in the range of 4000 mg/L. Seawater RO membranes can treat feed water with upper limit of 60,000 mg/L TDS (Greenlee et al., 2009). Therefore, RO was chosen as the primary desalination process. The desalination process (refer Fig. 4) started with pre-treatment and then was fed into a Low-Pressure RO (LPRO) which operates at 40 bars with a recovery of 90%, reducing TDS in the permeate (refer Table 2) around 600 mg/L. The LPRO reject, with a TDS of around 34,000 mg/L, is passed through a High-Pressure RO (HPRO) at 75 bar, achieving a further 80% recovery with TDS around 600 mg/L in permeated and reject TDS around 170,000 mg/L. Combined, the two-stage RO system delivered an overall recovery of 98%, minimizing reject volume and load on subsequent thermal treatment stage.

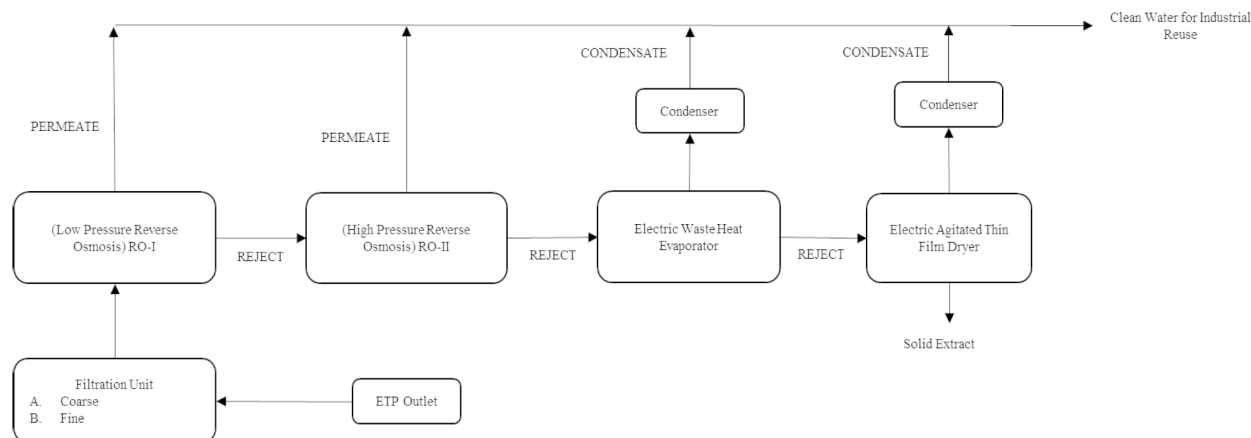


Figure 4—Process Flow Diagram of ZLD Desalination System.

Thermal Concentration. The reject from the HPRO stage exhibits extremely high salinity, with TDS levels exceeding 170,000 mg/L, which is well beyond the operational threshold of commercial RO membranes. To handle this concentrated brine, the process design succeeded with thermal evaporation stage due to the in-situ availability of 6MW waste heat with an average temperature gradient of 50°C. However, in the pilot trail, the HPRO reject was directed to an electric Waste Heat Evaporator (WHE) which gave a water recovery rate of around 85-90%. This also ensured flowability of the concentrated reject of brine slurry, which was further directed to an electric Agitated Thin Film Dryer (ATFD) for the extraction of solids while having a further water recovery of 35%. The whole thermal process averaged a vapor loss of around 0.2%, marking the completion of a near-ZLD cycle.

Desalinated water. The Feed water to ZLD desalinated system complied to 19 parameters out of 21 parameters as per regulatory standards (refer table 2). However, the desalinated water met all the 21 parameters (refer table 2) and was found suitable for inland discharge or internal re-use. The system demonstrated a remarkable water recovery of 99.975%, with minimal losses, 0.2% lost as vapor during the thermal stage and the remaining as traces of moisture in the solid residue.

Table 2—Comparative characterization of PFW at ETP Inlet, ETP Outlet and Desalination Output as per Regulatory Disposal Standards (CPCB, 2016).

Parameter	Discharge Standards not to exceed			ETP Inlet	ETP Outlet	Desalination Output	Test Method / Instrument Used
	On-shore Discharge	Marine Disposal	Subsurface Disposal**				
pH	5.5 – 9.0	5.5 – 9.0	—	6.5-8.5	7.5-8.4	5.5-7.3	Metrohm pH Meter
Temperature	40°C	—	—	38-45 °C	40-45 °C	—	—
Suspended Solids	100 mg/l	100 mg/l	100 mg/l	250-800 mg/l	40-80 mg/l	4-89 mg/l	Gravimetric method
Zinc	2 mg/l	0.1 mg/l*	—	BDL	BDL	NIL	UV-vis Spectrophotometer
BOD	30 mg/l	30 mg/l	—	262 mg/l	193 mg/l	26 mg/l	BOD Analyser
COD	100 mg/l	—	—	436 mg/l	353 mg/l	62 mg/l	UV-vis Spectrophotometer
Chlorides	600 mg/l	-	-	1376-2149 mg/l	1770-2345 mg/l	76-485 mg/l	Titration with silver nitrate
Sulphates	1000 mg/l	—	—	32 mg/l	26 mg/l	NIL	UV- vis Spectrophotometer

Parameter	Discharge Standards not to exceed			ETP Inlet	ETP Outlet	Desalination Output	Test Method / Instrument Used
	On-shore Discharge	Marine Disposal	Subsurface Disposal**				
TDS (Total Dissolved Solids)	2100 mg/l	-	-	2640-4720 mg/l	3000-4362 mg/l	167-618 mg/l	HACH Water Checker
% Sodium	60 mg/l	—	—	96.53 mg/l	<50 mg/l	47 mg/l	Na, K, Ca & Mg concentration
Oil and Grease	10 mg/l	10 mg/l	10 mg/l	10-1500 mg/l	<5 mg/l	0.0-7.6 mg/l	Horiba Oil Content Analyzer
Phenolics	1.2 mg/l	—	—	0.008 mg/l	<0.010 mg/l	0.04 mg/l	UV-vis Spectrophotometer
Cyanides	0.2 mg/l	0.005 mg/l *	—	BDL	BDL	NIL	UV-vis Spectrophotometer
Fluorides	1.5 mg/l	1.5 mg/l *	—	BDL	BDL	NIL	UV-vis Spectrophotometer
Sulphides	2.0 mg/l	—	—	BDL	BDL	NIL	UV-vis Spectrophotometer
Chromium (Cr ⁶⁺)	0.1 mg/l	—	—	BDL	BDL	NIL	UV-vis Spectrophotometer
Chromium (Total)	1.0 mg/l	0.1 mg/l *	—	BDL	BDL	0.1 mg/l	UV-vis Spectrophotometer
Copper	0.2 mg/l	0.05 mg/l *	—	BDL	BDL	NIL	UV-vis Spectrophotometer
Lead	0.1 mg/l	0.05 mg/l *	—	BDL	BDL	NIL	UV-vis Spectrophotometer
Mercury	0.01 mg/l	0.01 mg/l *	—	BDL	BDL	NM	—
Nickel	3.0 mg/l	0.1 mg/l *	—	BDL	BDL	NIL	UV-vis Spectrophotometer

*

To be measured at 50 m from the point of discharge

**

To be disposed at a depth of 1000 m below ground level

BDL Below Detection Limit, however least count satisfies regulatory standards

NM Not Measurable

Solid Extract. The pilot trail of ZLD Desalination system yielded around 3 kg of solid extract per kL of PFW. In India, solid waste is managed according to the Solid Waste Management Rules, 2016 notified under the Environment (Protection) Act, 1986. Conventionally, solid waste is disposed in engineered landfills, buried in lined pits to contain leachate. However, the salts present in the solid extract, being readily soluble in water, add complexity to the disposal process. In absence of a valorization strategy for the solids, the ZLD route effectively shifts the problem from liquid disposal to complex solid waste management. Recognizing this, OIL initiated R&D efforts to explore sustainable and value-added applications of the generated solid waste.



Figure 5—Images of PFW samples and solid extract: (a) Desalination inlet and desalination outlet water samples, (b) Solid extract generated from desalination process

Solid Extract Analysis

A 2 kg off-white color solid extract sample was collected and analyzed at the Chemical Department and the R&D Department of OIL for its composition and characteristics. A portion of the sample was ground to make homogeneous powder and tested using PetraMax XRF (Test Method: ASTM D4294) to check different elements present in the sample. On a part of the ground sample (homogeneous power) mineralogical analysis was carried out using powder X-Ray Diffractometer (XRD).

Table 3—XRD Analysis Report of Solid Extract from Desalination Plant at STF-Madhuban.

Sample Name	Minerals Quantified (%)				
	Halite	Gypsum	Sylvite	Natrite	Calcite
Solid Extract of Desalination Plant	54.3	1.1	0.3	39.0	5.3

From the analysis (refer Table 3 and Fig. 5), it was inferred that the sample contained mineral compounds, predominantly Halite (NaCl , 54.3%) and Natrite (Na_2CO_3 , 39.0%) with some amount of gypsum, and calcite. Also, the sample contained minor amounts of chloride, sulphate and carbonates.

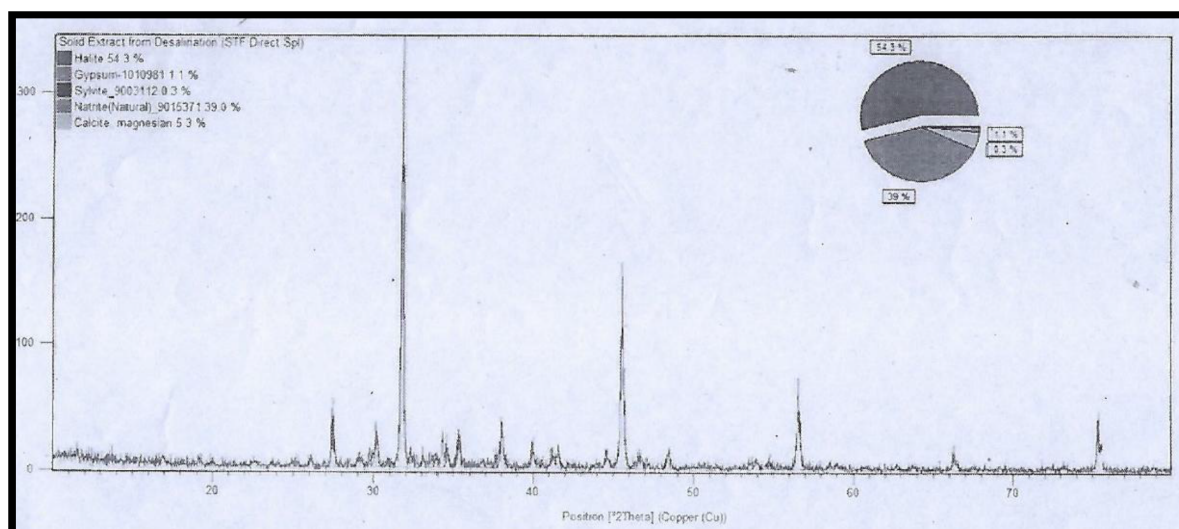


Figure 5a—XRD Spectrum of solid extract showing intensity (counts) versus angle 2θ .

Solid Extract Utilization

Brine Preparation. The solid extract analysis revealed the major constituent as Halite and Natrite. Presently salt like Halite is being utilized for preparation of brine solutions for workover operations in oil and gas industry. Similar usages of the solid extract generated from the desalination system were explored. The initial trial of brine preparation using the solid extract resulted in pink solution forming and precipitate formation. The further attempt to utilize the solid extract for similar usage was dropped.

Market Survey. A market survey was carried out to explore the potential users of the solid extract to produce value-added products or utilize the extract in a useful manner. Through the market survey it was found that halite and natrite were widely used in various industries. Halite is used in the food industry, chemical industry, water softening, etc. Natrite is used in detergents and soaps, water treatment, paper production, etc. Since the solid extract consists of halite and natrite in majority, their separation is complicated and the market unwilling to utilize the generated solid extract on their own.

Valorization of Solid Extract. OIL initiated its own research and development program to explore the utilization of solid extract generated from the ZLD process. The objective was to convert this waste stream into value-added or commercially viable products. This initiative was inspired by the well-established model of fly ash utilization from thermal power plants, which is extensively used in cement manufacturing, road construction, and other building material applications. Experimental trials were conducted to develop building materials by blending the solid extract with binders, additives and stabilizing agents. Initial laboratory testing yielded promising results, with one formulation, Sample D, exhibiting the highest compressive strength among all tested compositions. This formulation contained 35% solid extract by weight (refer Table 4 and Fig. 6).

Table 4—Lab study of solid extract utilization, samples prepared using solid extract and other constituents in lab.

Constituents	Sample B	Sample E	Sample D
Solid extract (%)	25	30	35
Constituent A* (%)	30	30	30
Constituent B* + Additives* (%)	45	40	30

*

Chemical composition of constituents and additives not disclosed

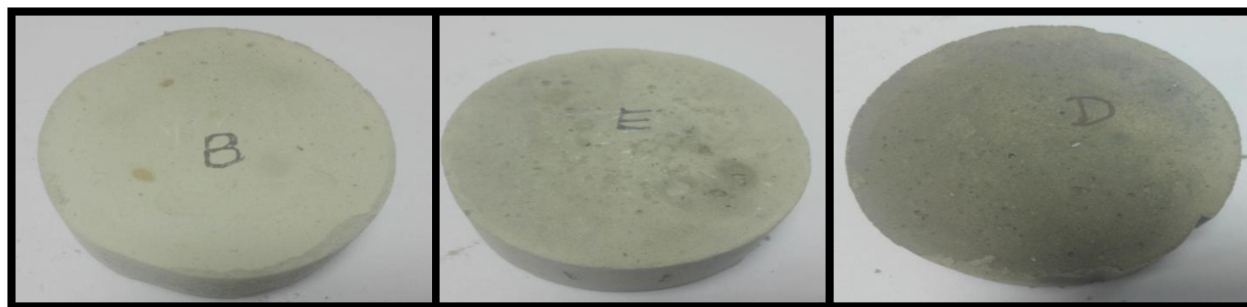


Figure 6—Images of samples prepared using solid extract and other constituents in lab.

Subsequent formulations, based on Sample D, were developed into building material prototypes and subjected to mechanical strength testing. Uniaxial compressive strength (UCS) tests were conducted using a 250 kN Servo Hydraulic Universal Testing Machine (UTM) [Make: BISS MEDIAN 250] at a strain rate of 1 mm/min. The unconfined compressive strength (UCS) values recorded for samples SB1, SB2, SB3, and SB4 were 8.70 MPa, 4.41 MPa, 6.81 MPa, and 7.28 MPa, respectively. These values fall within the typical strength range of BIS standard burnt clay bricks (IS 1077:1992) for respective applications, indicating the solid extract's viability as a structural component in construction materials (refer Fig. 7)

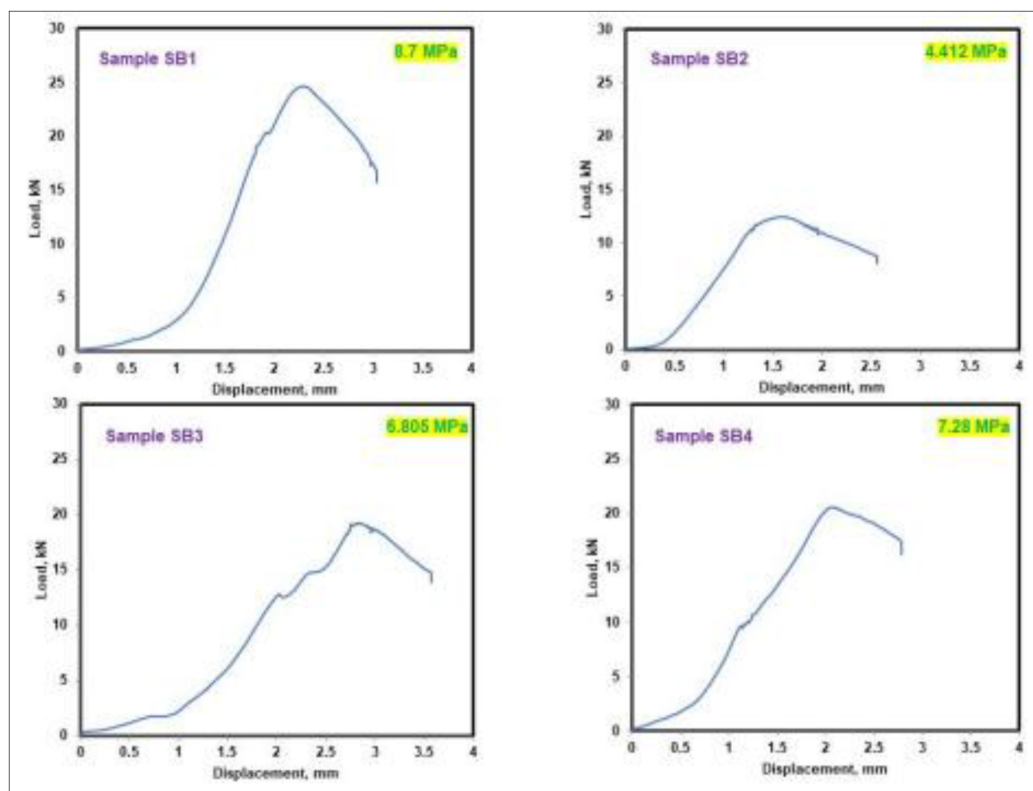


Figure 7—Universal Compressive Strength Analysis of building material made from ZLD desalination solid extract.

The certification of the formulated construction materials, as per BIS standards, is currently in process. Nevertheless, the positive results mark a pivotal advancement toward the effective valorization of desalination derived solid waste, transforming it from a disposal liability into a potential value-added and sustainable product.

In a similar manner to construction materials, garden planters and pots are also being experimented with varying composition of solid extract, additives and stabilizers.

Conclusion and Discussion

Combining PFW treatment in desalination system with solid waste valorization provides a new perspective on the management of PFW. Instead of treating PFW as the largest byproduct (Viel and Clark, 2011) and liability of the oil and gas sector, the novel methodology converts the PFW into water suitable for inland disposal or industrial reuse which is a valuable product. Along with this, the solid extract generated can further be valorized using cutting-edge scientific research, development and technologies. Traditional liability can be converted into dual revenue stream through value added products. The proposed methodology not only contributes to solving the long-term problem of oil and gas sector of PFW handling but also contributes to water conservation, through desalinated water inland surface dispose or internal re-use. The desalinated water can be used for industrial purposes, agricultural purposes, recharging underground aquifer, etc. Further similar methodology can be exploited in the desalination process of sea water for generation of reusable water where instead of brine disposal in marine bodies, solid extract can be valorized and protect marine bodies from contamination.

PFW is not a universal commodity as its composition and properties vary across reservoirs. The pilot trial and valorization of solid extract carried out at STF-Madhuban of OIL may not be replicable as it is highly dependent on the PFW composition. However, similar procedures can be used in the oil and gas sector with customized valorization of solid extract instead of disposal. Further study may be carried out to derive a universal methodology for solid extract valorization despite varying composition of solid extract to make this methodology more accessible in industry.

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Nomenclature

PFW	: Produced formation water
OIL	: Oil India Limited
ZLD	: Zero Liquid Discharge
RO	: Reverse Osmosis
ETP	: Effluent Treatment Plant
LPRO	: Low-Pressure Reverse Osmosis
HPRO	: High-Pressure Reverse Osmosis
R&D	: Research and Development
KLDP	: Kilo Liters Per Day
TDS	: Total Dissolved Solids
EOR	: Enhanced Oil Recovery
TSDF	: Treatment Storage and Disposal Facilities
BTEX	: Benzene, Toluene, Ethylbenzene, and Xylene
COD	: Chemical Oxygen Demand
PAH	: Polycyclic Aromatic Hydrocarbons
BIS	: Bureau of India Standards
CPCB	: Central Pollution Control Board
TSS	: Total Suspended Solids
BOD	: Biochemical Oxygen Demand

CTPI : Corrugated Tilted Plate Interceptor
 DAF : Dissolved Air Flotation
 MGF : Multigrade Filter
 ORF : Oil Removal Filter
 UF : Ultra-Filtration
 STF : Secondary Tank Farm
 WHE : Waster Heat Evaporator
 ATFD : Agitated Thin Film Dryer
 XRD : X-Ray Diffractometer
 UCS : Uniaxial Compressive Strength
 UTM : Universal Testing Machine
 kL : Kilo Liters
 IS : Indian Standard

Conversion Factor

Kilo Liters Per Day : 1 kLpd = 6.2898 bpd
 (kLpd) to Barrel Per
 Day (bpd)

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